

2.6 Motion Along a Curve Worksheet

1. Find the velocity and position vectors of a particle whose acceleration is given by $\mathbf{a}(t) = \langle 2t, \sin(t), \cos(2t) \rangle$ with initial velocity $\mathbf{v}(0) = \mathbf{i}$ and initial position $\mathbf{r}(0) = \mathbf{j}$.

We know that $\mathbf{v}(t) = \int \mathbf{a}(t) dt = \left\langle t^2, -\cos(t), \frac{1}{2} \sin(2t) \right\rangle + \mathbf{C}_1$.

Since we know that $\mathbf{v}(0) = \mathbf{i} = \langle 1, 0, 0 \rangle$ then $\langle 1, 0, 0 \rangle = \langle 0, -1, 0 \rangle + \mathbf{C}_1 \Rightarrow \mathbf{C}_1 = \langle 1, 1, 0 \rangle$.

Therefore, $\mathbf{v}(t) = \left\langle t^2 + 1, 1 - \cos(t), \frac{1}{2} \sin(2t) \right\rangle$.

We then know that $\mathbf{r}(t) = \int \mathbf{v}(t) dt = \left\langle \frac{1}{3}t^3 + t, t - \sin(t), -\frac{1}{4} \cos(2t) \right\rangle + \mathbf{C}_2$

Since we know that $\mathbf{r}(0) = \mathbf{j} = \langle 0, 1, 0 \rangle$ then $\langle 0, 1, 0 \rangle = \left\langle 0, 0, -\frac{1}{4} \right\rangle + \mathbf{C}_2 \Rightarrow \mathbf{C}_2 = \left\langle 0, 1, \frac{1}{4} \right\rangle$.

Therefore, $\mathbf{r}(t) = \left\langle \frac{1}{3}t^3 + t, t - \sin(t) + 1, \frac{1}{4} - \frac{1}{4} \cos(2t) \right\rangle$.

2. The position function of a particle is given by $\mathbf{r}(t) = \langle t^2 - 9t + 12, 4t + 5, 3t + 8 \rangle$. At what time is the particle travelling at its minimum speed?

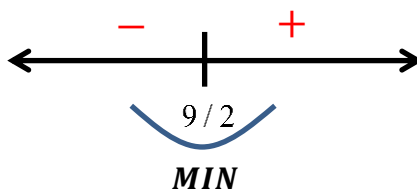
We know that the velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t) = \langle 2t - 9, 4, 3 \rangle$. This implies that the speed of the particle is given by $v(t) = \|\mathbf{v}(t)\| = \sqrt{(2t - 9)^2 + 16 + 9} = \sqrt{4t^2 - 36t + 106}$.

We want to minimize the function $v(t)$ and we can do this using our knowledge from Calculus I.

To minimize $\sqrt{f(x)}$ it suffices to minimize just $f(x)$.

With this in mind we calculate:

$$f'(t) = 8t - 36 \text{ and } f'(t) = 0 \text{ when } 8t - 36 = 0 \Rightarrow t = \frac{9}{2}$$



We see that the particle reaches its minimum speed at $t = \frac{9}{2}$.

3. Suppose a projectile is fired with an initial speed of 60 m/s at an angle of 45° from the initial position of $\langle 1, 1 \rangle$.
- a) Write a vector function which will compute the position of the projectile at any time t .

We know that $\mathbf{r}(t) = \langle v_0 \cos(\alpha)t + a, v_0 \sin(\alpha)t - 4.9t^2 + b \rangle$ models the motion of the projectile.

Therefore, the needed function would be:

$$\mathbf{r}(t) = \langle 60 \cos(45^\circ)t + 1, 60 \sin(45^\circ)t - 4.9t^2 + 1 \rangle = \langle 30\sqrt{2}t + 1, 30\sqrt{2}t - 4.9t^2 + 1 \rangle$$

- b) At what time will the projectile hit the ground?

We know the projectile will hit the ground when its height reaches a value of 0. Therefore we solve:

$$30\sqrt{2}t - 4.9t^2 + 1 = 0 \rightarrow t = \frac{-30\sqrt{2} \pm \sqrt{(30\sqrt{2})^2 - 4(-4.9)(1)}}{2(-4.9)} = \frac{30\sqrt{2} \pm \sqrt{1819.6}}{9.8}$$

This implies that $t \approx 8.68 \text{ sec}$ or $t \approx -0.02 \text{ sec}$

This obviously means that the particle hits the ground after $t = \frac{30\sqrt{2} + \sqrt{1819.6}}{9.8} \text{ sec} \approx 8.68 \text{ sec}$.

- c) At what time will the projectile reach its maximum height?

We know that when the projectile reaches its maximum height its velocity in the vertical direction must equal 0. Therefore, we compute $\mathbf{v}(t) = \langle 30\sqrt{2}, 30\sqrt{2} - 9.8t \rangle$ and solve:

$$30\sqrt{2} - 9.8t = 0 \rightarrow t = \frac{30\sqrt{2}}{9.8} \approx 4.33 \text{ sec}$$

- d) Estimate the maximum height obtained by the projectile (round your answer to 2 decimals).

The maximum height obtained by the projectile would occur at the value of t from part (c). Therefore, we compute:

$$30\sqrt{2} \left(\frac{30\sqrt{2}}{9.8} \right) - 4.9 \left(\frac{30\sqrt{2}}{9.8} \right)^2 + 1 \approx 92.84 \text{ meters}$$

- e) At what time will the projectile be 100 meters downrange?

The projectile will be 100 meters downrange when the x -component of $\mathbf{r}(t)$ equals to 100. So, we compute:

$$30\sqrt{2}t + 1 = 100 \rightarrow t = \frac{99}{30\sqrt{2}} = \frac{33}{10\sqrt{2}} \text{ sec} \approx 2.33 \text{ sec}$$

- f) Estimate the height of the projectile when it is 200 meters downrange (round your answer to 2 decimals).

The projectile will be 200 meters downrange when the x -component of $\mathbf{r}(t)$ equals to 200. So, we compute:

$$30\sqrt{2}t + 1 = 200 \rightarrow t = \frac{199}{30\sqrt{2}} \text{ sec} \approx 4.69 \text{ sec}$$

So, the height of the projectile at this time is given by the y -component of $\mathbf{r}(t)$. So, we compute:

$$30\sqrt{2} \left(\frac{199}{30\sqrt{2}} \right) - 4.9 \left(\frac{199}{30\sqrt{2}} \right)^2 + 1 \approx 92.20 \text{ meters}$$

- g) Suppose we could change the angle at which the projectile is fired. What angle could we fire the projectile at so that it would be 100 meters downrange after 10 seconds?

If we plan to change the angle the projectile is fired at, then we must adjust our original equation to be:

$$\mathbf{r}(t) = \langle 60 \cos(\alpha)t + 1, 60 \sin(\alpha)t - 4.9t^2 + 1 \rangle$$

If we want the projectile to be 100 meters downrange after 10 seconds then we use the x -component to solve for the angle α under these conditions.

$$60 \cos(\alpha)(10) + 1 = 100 \rightarrow \cos(\alpha) = \frac{33}{200} \quad (\text{We should note that there are two angles between}$$

0 and 2π for which the cosine could equal $\frac{33}{200}$, however we only want the acute angle since we want to ensure that we are indeed shooting the projectile *downrange*! Safety first!)

$$\text{Therefore, } \cos(\alpha) = \frac{33}{200} \rightarrow \alpha = \cos^{-1} \left(\frac{33}{200} \right) \approx 1.41 \text{ radians or } \approx 80.5^\circ$$

4. Suppose that the velocity and acceleration vectors are perpendicular for the given curve $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$. Then what can you say about the speed of this curve?

We know that the velocity vector is given by: $\mathbf{v}(t) = \mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$

We know that the acceleration vector is given by: $\mathbf{a}(t) = \mathbf{r}''(t) = \langle x''(t), y''(t), z''(t) \rangle$

If these vectors are perpendicular, then this means that their dot product equals 0. This means:

$$x'(t)x''(t) + y'(t)y''(t) + z'(t)z''(t) = 0$$

We may observe that if we multiply both sides by 2, then the equation will resemble the derivative of the following equation:

$$x'x'' + y'y'' + z'z'' = 0 \Leftrightarrow 2x'x'' + 2y'y'' + 2z'z'' = 0$$

$$\Leftrightarrow \frac{d}{dt}[(x')^2 + (y')^2 + (z')^2] = \frac{d}{dt}[\text{constant}]$$

$$\Leftrightarrow (x')^2 + (y')^2 + (z')^2 = \text{constant}$$

$$\Leftrightarrow \sqrt{(x')^2 + (y')^2 + (z')^2} = \text{constant}$$

However, also notice that this is the formula to calculate the speed of the curve! Therefore, we can conclude that the speed of the curve must be constant!